DSCI 505 — Data Visualization for Data Science

Homework 1: Seeing & Designing Data — Written Report

Ahmad Avar Naggayev · June 29, 2026 · Companion to: DSCI505_HW1_Ahmad_Naggayev.ipynb

Part A — Week 1: Why We Visualize, Cognition & Tools

A1 — Anscombe's Quartet: The Case for Charts

What the summary statistics show: All four datasets have virtually identical means (x■≈9, ■≈7.5), variances (~11 and ~4.1), correlations (~0.816), and regression lines (y ≈ 3 + 0.5x). Looking only at numbers, you would conclude the datasets are interchangeable.

What the pictures reveal: Dataset I is a well-behaved linear relationship with random noise. Dataset II is a clear quadratic curve — fitting a line is a modeling error. Dataset III is near-perfect linear with one high-leverage outlier pulling the regression slope; invisible in the stats. Dataset IV has all x-values = 8 except one outlier at x=19 — the entire regression line is determined by a single pathological point.

Lesson: Summary statistics are a compression — they discard spatial arrangement that reveals structure, outliers, non-linearity, and leverage. In any data science workflow, exploratory visualization is not optional — it is the first check that your statistical model is even appropriate for the data you have.

A2 — Perception & Ranking of Encodings

Encoding ranking (most → least accurate): (1) Position on a common scale — most accurate; (2) Length; (3) Angle/slope; (4) Area; (5) Color saturation; (6) Color hue — least accurate for quantitative values (nominal channel).

Preattentive attributes are visual features the brain processes in parallel before conscious attention (~200ms), regardless of display size. Examples: (1) Color — a red dot among grey dots is found instantly via parallel color processing. (2) Orientation — a tilted line among vertical lines pops out immediately. A single '5' among '3's is NOT preattentive because digit identity requires serial higher-level pattern recognition: each character must be parsed sequentially, taking time proportional to the number of distractors.

Best encoding per task: (a) Comparing 8 regions → sorted bar chart (position on common scale); (b) Share of a whole → sorted bar with % (position more accurate than pie angles); (c) Correlation between two variables → scatter plot (both axes use position, directly encoding the bivariate relationship).

A3 — The Tooling Landscape

Tool	Use Case	Static/Interactive	Learning Curve	Strength / Weakness
matplotlib	Publication figures, custom control	Static	Medium	Max flexibility / Verbose boilerplate
seaborn	Statistical EDA, tidy data	Static	Low	Beautiful defaults / Hard to customize deeply
plotly	Interactive dashboards, web	Interactive	Medium	Hover/zoom built-in / Heavy for simple charts
Tableau	Business BI, non-technical	Interactive	Low–Med	Drag-and-drop / Expensive, no prog. control
Excel	Ad-hoc quick charts	Mostly static	Very low	Ubiquitous / No reproducibility

A4 — First Plot, AI-Assisted (Prompt + Critique)

Prompt used:

```
"Load sales.csv and produce a fully-labelled exploratory plot that shows monthly revenue trend
from 2022–2024, broken down by region. Choose an appropriate chart type and explain your choice."
```

What the AI got right: Chose a line chart over time — appropriate for a continuous numeric quantity over ordered time intervals. x-axis is ordered time; one line per region; legend present.

What I had to fix (4 corrections): (1) AI used the default matplotlib color cycle which is not colorblind-safe — replaced with Okabe-Ito subset. (2) y-axis was unlabelled in raw AI output — added 'Revenue (USD)' with dollar formatting. (3) AI did not format y-axis as currency — large values like 100000 read poorly; added `${x:,.0f}` formatting. (4) Month aggregation used `.resample('M')` which triggers a deprecation warning on newer pandas — switched to `.dt.to_period('M')`.

Part B — Week 2: Principles of Effective Visualization & Color

B1 — CRAP Critique & Redesign

Four problems named: (1) **Contrast (color):** Four unrelated rainbow colors create false visual hierarchy — the eye is drawn to lime-green and magenta because they're perceptually loudest, not because those categories matter more. (2) **Alignment:** Bars are unsorted; every pairwise comparison requires scanning the whole chart. (3) **Data-ink (Tufte):** The thick black grid is chartjunk — heavy lines carrying zero data. (4) **Contrast / Proximity:** 'REVENUE!!!' is not a title — it names the variable already on the axis (redundant) while missing the question the chart answers. No y-axis label or units.

Redesign decisions: Horizontal bar chart (long labels read left-to-right), sorted ascending (largest at top for easy reading), single blue (color encodes nothing here), direct dollar value labels eliminating axis-tracing, minimal grid, proper question-based title, dollar-formatted axis with units.

B2 — Color, Done Right

Three palette choices: (1) **Categorical** → **Okabe-Ito** for Region — each color encodes a distinct nominal category; Okabe-Ito is colorblind-safe and perceptually equidistant. (2) **Sequential** → **'Blues'** for revenue magnitude — all values are positive; darker = more revenue, lighter = less; monotonic luminance matches monotonic data. (3) **Diverging** → **'RdBu_r'** for correlation — correlations range from –1 to +1 with 0 as a natural center; red = positive, blue = negative, white = near-zero.

Jet vs. viridis: The `jet` colormap has non-monotonic luminance — yellow and cyan are perceptually much brighter than red and blue, creating *false edges* at those hue transitions that look like real data features. `viridis` has monotonically increasing luminance (dark-to-light as values rise), so every perceptual difference corresponds to a real data difference.

Checking colorblind safety (Okabe-Ito): Convert the figure to simulated CVD using `colorspacious.cspace_convert`, or upload the PNG to a CVD simulator (e.g. Coblis). Alternatively: print in grayscale — if all categories remain distinguishable, the palette is robust. Okabe-Ito is specifically designed to remain distinguishable under all three common forms of CVD (deuteranopia, protanopia, tritanopia).

B3 — Data-Ink Ratio & Chartjunk

Tufte's data-ink ratio: The fraction of a chart's total ink devoted to non-redundant data. Principle: maximize the share of ink that encodes actual data; erase everything else.

Removed from the bad chart: Grey background fill, white gridlines on grey (redundant), legend repeating x-axis tick labels, multi-color bars encoding nothing, thick black bar outlines. **How removing ink increases information:** The direct value labels in the redesign convey exact dollar amounts in the same space the legend previously wasted — less total ink, more information per mark.

B4 — Accessibility Recolor, AI-Assisted (Prompt + Critique)

Prompt used:

```
"Take this regional revenue bar chart which uses matplotlib's default color cycle. Recolor it to
be colorblind-safe (red-green CVD) and also readable in grayscale. Explain which palette you
chose and why."
```

What the AI got right: Correctly recommended Okabe-Ito, which was designed by vision researchers specifically for CVD safety. Correctly identified the red-green pair in the default cycle (`#2ca02c` green and `#d62728` red) as the primary CVD problem for ~8% of men with deuteranopia.

What hurt clarity: The Okabe-Ito orange and sky blue are very different in color but closer in luminance than the default palette — in strict grayscale the bars are harder to distinguish by shade alone (luminance gap ~0.14 vs. ~0.25 for the default). For a chart where color is the *only* differentiator with no text labels, I would add hatch patterns or direct bar labels as a secondary encoding to ensure grayscale readability is not solely dependent on subtle luminance differences.

Part C — Week 3: Chart & Graph Fundamentals

C1 — Chart Choice Table

#	Question	Best Chart	Justification
1	Monthly revenue over 3 yrs	Line chart	Continuous time x-axis; quantity over ordered time — line shows trend
2	Which category sells most	Bar chart	Nominal categories on single magnitude; position = most accurate
3	Distribution of order values	Histogram	Shape of continuous numeric distribution — spread, skew, modality
4	Share of revenue by region	Sorted bar (not pie)	Position more accurate than pie angles for 4 similar-size slices
5	Regions compared on revenue	Bar chart	Comparing totals across categories; common zero baseline
6	Units vs. revenue relationship	Scatter plot	Two continuous vars; bar fails (requires aggregation, loses variation); line implies time sequence

C2 — Bars Done Right & How They Lie

Truncated axis distortion: The misleading chart starts the y-axis at ~$1.55M. The difference between West and South looks visually enormous (~4× bar height) when it is actually ~10% in real terms. Bar charts encode data in *length from the baseline*; cutting the baseline off at a non-zero value makes small absolute differences look like massive ratios.

Sorting aids comparison: An unsorted chart forces the reader to scan all bars and make pairwise comparisons by eye. A sorted chart makes rank immediately visible — the eye compares adjacent bars without scanning.

Horizontal bars: When category labels are long (multi-word text, annotations), horizontal bars let labels read left-to-right naturally without 45° rotation — slow reading, head-tilting, and truncation all disappear.

C3 — Histograms & Binning

Effect of bin width: 5 bins hides all shape — just a big block. 30 bins reveals the right skew and long upper tail from high-value Technology/Furniture orders. 100 bins buries signal in noise — individual bin counts fluctuate randomly, creating spikes that look like structure but are sampling variability.

Skew: Right-skewed (positive) — most orders cluster in lower revenue, with a long tail from Technology orders (~$600 base price). Mean > Median confirms right skew.

When a box plot is better: When comparing revenue *across groups* (by region or category). A single histogram hides whether skew is uniform across regions or driven by one. Box plots show median, IQR, and outliers for each group simultaneously.

C4 — The Pie-Chart Problem

Why bar chart is easier to read: Pie charts encode data in arc angle and area — both channels humans read poorly, especially near 25% (90°) or when comparing non-adjacent slices. The bar chart uses position on a common zero baseline: any two bars compare accurately by reading their heights.

Narrow conditions where a pie is acceptable: (1) 2–3 slices maximum, (2) the comparison is a simple majority / 'more than half' message, not a precise comparison, (3) the audience is non-technical and the rough part-of-whole message is the entire point. Example: '68% voted yes' — a two-slice pie conveys that message instantly.

C5 — Does the AI Pick the Right Chart? (AI-Assisted)

Question 1 tested (C1 #3):

```
"What chart should I use to show the distribution of individual order values from a retail
dataset?"
```

AI answer: Use a histogram. Shows frequency distribution of a continuous numeric variable. Can overlay a KDE curve for a smoother view. **Verdict: Correct.** AI correctly identified the right chart. Where it could steer a novice wrong: it did not warn that bin width choice dramatically changes the apparent story (5 vs. 30 vs. 100 bins). A novice might accept the default bin count as the truth.

Question 2 tested (C1 #4):

```
"What chart should I use to show what share of total revenue comes from each of four sales
regions?"
```

AI answer: A pie chart is the classic choice for showing part-of-whole data. **Verdict: Partially wrong.** AI defaulted to the culturally expected answer without mentioning that bar charts are preferred by visualization researchers for precisely this task. Four slices of similar size is the hardest possible case for angle comparison. A sorted bar with a common baseline is more accurate to read, requires no mental angle estimation, and makes exact values easier to compare. The AI would steer a novice into a pie chart that actively obscures the differences it's meant to reveal.

Part D — Week 4: Advanced Graph Types & Multidimensional Data

D1 — Scatter Plots & Multidimensional Encoding

Four dimensions encoded: x = marketing spend (position), y = revenue (position), color = region (categorical — Okabe-Ito), marker size = units ordered (magnitude).

Correlation (marketing spend vs. revenue): r ≈ 0.97 — mechanically high because marketing spend was generated as 8% of revenue, so the relationship is structural rather than causal. Reported in the chart title so a reader is not misled into a causal interpretation.

Overplotting remedy: Setting alpha=0.25 reveals density — dark clusters show where orders concentrate; at opacity=1 every high-density region looks identical to a single isolated point, hiding all distributional information.

D2 — Box Plots & Distributions by Group

Box annotations: The box spans Q1–Q3 (IQR — middle 50% of orders). The line inside is the median (Q2). Whiskers extend to the farthest point within 1.5×IQR from each quartile. Points beyond the whiskers are individual outliers.

What the violin reveals that the box hides: The full shape of the distribution — bimodality, skew, flat shoulders, and long tails. Two groups with identical Q1/Q2/Q3 can have completely different distribution shapes; a box hides this, a violin exposes it.

Median vs. total: The region with the highest median order value (West, 1.25× regional multiplier) is not the same as the highest total revenue region — total depends on both per-order value AND number of orders. North has the highest order share (30%) so it can generate more total revenue at a lower per-order multiplier. Median and total answer fundamentally different business questions.

D3 — Heat Maps & Colormap Choice

Correlation matrix → Diverging ('RdBu_r'): Correlations are centered on 0; a diverging map makes negative (blue) and positive (red) values visually distinct. A sequential map would make –0.9 and +0.9 look similar (both far from zero) when they are opposites.

Region × Month revenue → Sequential ('Blues'): Revenue is always non-negative — there is no zero midpoint to diverge around. Darker = more revenue.

Common pitfall noted: Using a diverging map on all-positive revenue data centers white at the mean revenue value, making average months look 'neutral' or near-zero — this creates false structure implying some cells represent negative revenue when they do not.

D4 — Area Charts

What each chart answers: (1) Stacked area (absolute) → 'What is the total month by month and how does each category contribute?' (2) 100% stacked → 'Is the revenue mix shifting over time?' (3) Plain lines → 'How does each individual category trend?' Only the lines chart lets you accurately compare individual category trajectories because each has its own baseline at zero.

When stacked area misleads: Middle bands have neither a flat baseline nor a common reference — their top and bottom edges both move. Reading a middle band's height requires visually subtracting two curved lines, which humans do poorly. A category might appear flat when it is actually growing, pushed up by a rising band below it. **Alternative:** Small multiples (one line chart per category) or indexed lines (all starting at 100).

D5 — One-Screen Dashboard, AI-Assisted (Prompt + Critique)

Prompt used:

```
"Using the sales.csv dataset, create a 2×2 multi-panel dashboard that tells one coherent story
about this retail business: (1) overall revenue trend, (2) regional comparison, (3) order value
distribution, (4) marketing spend vs. revenue relationship. Use consistent color for region,
label all axes with units, and make it suitable for a business review meeting."
```

Panel-by-panel audit and fixes:

Panel 1 (Revenue Trend): Chart type correct — bar + rolling average line. AI produced only the bar chart without a rolling average, missing the trend signal. **Fixed: added 3-month rolling average line.**

Panel 2 (Regional comparison): Horizontal sorted bar correct. AI used the default color cycle, breaking region-color consistency with Panel 4 (same regions appeared different colors). **Fixed: applied Okabe-Ito region palette consistently across all panels.**

Panel 3 (Order distribution): Histogram correct. AI omitted median and mean reference lines — the right-skewed distribution has mean > median, which is a key business insight (a few large Technology orders pull the average up). **Fixed: added both reference lines with dollar-formatted annotations.**

Panel 4 (Marketing vs. Revenue): Scatter correct. AI used alpha=1.0, causing severe overplotting on 2,000 points — density invisible. Also: the high correlation (r≈0.97) is structural (marketing = 8% of revenue by construction), not a causal finding — a reader could incorrectly conclude 'spend more on marketing → more revenue.' **Fixed: alpha=0.2 for overplotting; added correlation context note to title.**

Reflection

Of every chart I made, the Anscombe's Quartet 2×2 scatter panel (A1) communicates its message most effectively — four scatter plots with identical summary statistics but completely different shapes is one of the most striking arguments for visualization that exists, and it requires no annotation to land. The most practically useful skill I built is the CRAP framework (B1): naming concrete principles turns vague intuitions about 'bad charts' into actionable edits, and knowing which principle a design choice violates makes redesign systematic rather than aesthetic. The most commonly violated principle I observed in both the homework examples and in every AI-generated draft was data-ink ratio: every AI output had non-data elements — legend redundancy, default gridlines, unlabelled axes — that required cleanup before the chart was ready to show anyone. Colormap choice (B2) surprised me most: the jet vs. viridis luminance plot made the perceptual harm of rainbow maps concrete in a way that 'rainbow maps are bad' never did.